

Notes: Schularick and Taylor (AER, 2012)
Credit Booms Gone Bust: Monetary Policy, Leverage Cycles, and Financial
Crises, 1870–2008

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Data and measurement	3
3.1	Country coverage and time period	3
3.2	Core financial variables	3
3.3	Macroeconomic and asset-price variables	4
3.4	Financial-crisis dates	4
4	Methodology and empirical specifications	5
4.1	Long-run trends and year effects	5
4.2	Event analysis after financial crises	5
4.3	Financial-crisis prediction model	5
4.4	Money versus credit	6
4.5	Robustness and additional predictors	6
5	Identification and interpretation	7
5.1	What the paper can identify	7
5.2	Why the credit variable is informative	7
5.3	Why country fixed effects matter	7
5.4	Forecasting versus explanation	7
5.5	Postwar policy interpretation	8
5.6	Main interpretation of the paper	8
6	Tables and figures	8
6.1	Table 1 and Figure 1: Summary statistics and the rise of financial aggregates	8
6.2	Figures 2 and 3: Credit decouples from broad money	9
6.3	Figures 4 and 5: Financial and real variables after crises	10
6.4	Figure 6 and Table 2: Inflation and cumulative crisis effects	10

6.5	Table 3 and Figure 7: Baseline crisis prediction	11
6.6	Tables 4 and 5: Credit versus money, and prewar versus postwar samples	12
6.7	Figure 8 and Table 6: Predictive comparisons and robustness checks	13
6.8	Tables 7 and 8: Asset prices, financial development, and interactions	14
7	Short summary	15
8	Conclusion	16
8.1	Core conclusion	16
8.2	Economic intuition	16
8.3	Incremental contribution revisited	16
8.4	Implications	16
8.5	Limitations	16
8.6	Takeaway	16

1 Big picture

- **Research question.** What do long-run data say about money, bank credit, leverage, monetary policy, and financial crises? More specifically, are financial crises better predicted by monetary aggregates or by bank-credit booms?
- **Main idea.** The paper builds a long historical panel for 14 advanced economies from 1870 to 2008. It uses this panel to show that credit and money behaved similarly before World War II, but diverged strongly after World War II as banking systems became more leveraged and relied increasingly on nonmonetary liabilities.
- **Empirical object.** The paper studies three objects: long-run trends in money and credit, event-study dynamics after banking crises, and early-warning regressions that predict financial crises using lagged credit and money growth.
- **Main finding.** Credit growth is a robust predictor of future financial crises. Broad and narrow money growth add much less predictive content, especially in the postwar period.
- **Two eras of finance capitalism.** In the prewar era, money and credit moved closely together. In the postwar era, bank loans and bank assets rose much faster than broad money and GDP, so credit became a separate macro-financial state variable rather than merely the asset-side mirror of money.
- **Crisis dynamics.** After World War II, central banks and governments prevented collapses in money and deflation after crises, but crises still produced large contractions in real GDP, real investment, bank loans, and bank assets.
- **Policy message.** Monetary policy that watches only inflation and the output gap misses an important warning signal. Credit booms and leverage cycles deserve attention in macroprudential and central-bank policy.

Economic takeaway. The paper moves the discussion from a pure *money view* toward a *credit view*. Financial instability is not just about movements in the money stock; it is about balance-sheet expansion, leverage, and rapid credit creation that can later reverse sharply.

2 Incremental contribution of the paper

- **A new long-run credit dataset.** The paper assembles annual bank loan, bank asset, monetary, macroeconomic, stock price, and crisis data for 14 advanced economies over almost 140 years. This is much broader in both time and cross-country coverage than the typical postwar macro-finance dataset.
- **Separates money from credit.** Earlier monetary history often treats the banking system mainly through its monetary liabilities. This paper explicitly compares money aggregates with bank loans and bank assets, showing that they are no longer interchangeable after World War II.
- **Documents the financialization trend.** The paper shows that postwar banking systems grew rapidly relative to GDP and, even more importantly, relative to broad money. This makes leverage and nonmonetary bank funding central to the analysis.

- **Compares crisis regimes.** The event analysis contrasts prewar and postwar financial crises. The comparison shows that the postwar regime avoided the classic money collapse and debt deflation, but not the real output costs of crises.
- **Early-warning evidence.** The paper estimates predictive models of banking crises using lagged macro-financial variables. Lagged real credit growth performs well; money growth does not.
- **Links historical macroeconomics to modern policy debates.** The results speak directly to post-2008 questions about whether central banks should monitor credit booms, asset-price booms, and leverage, rather than only inflation and real activity.
- **Long-horizon perspective on rare events.** Financial crises are rare in any single country sample. The paper's long panel creates enough crisis observations to study patterns that would be hard to detect in short postwar data.

3 Data and measurement

3.1 Country coverage and time period

The paper builds an annual panel for 14 developed economies over 1870–2008.

- **Countries.** The sample covers the United States, Canada, Australia, Denmark, France, Germany, Italy, Japan, the Netherlands, Norway, Spain, Sweden, Switzerland, and the United Kingdom.
- **Years.** The headline sample spans 1870–2008, with war and immediate aftermath periods excluded in many summary statistics and regressions.
- **Two main regimes.** The authors emphasize a pre-World War II era, roughly 1870–1939, and a post-World War II era, roughly 1945–2008.
- **Why history matters.** Financial crises are rare events. A short postwar sample has too few crises to separate normal cyclical movements from crisis dynamics and early-warning signals.

Interpretation. The dataset is designed for macro-financial history. It is not a micro dataset about individual banks or borrowers. Its strength is that it allows broad, long-run comparisons of credit, money, and crises across many advanced economies.

3.2 Core financial variables

The key financial aggregates are bank loans, bank assets, narrow money, and broad money.

- **Bank loans.** Total lending is the end-of-year stock of outstanding domestic-currency lending by domestic banks to domestic households and nonfinancial corporations. Lending within the financial system is excluded.
- **Bank assets.** Total bank assets are the end-of-year balance-sheet assets of banks with national residency, excluding foreign-currency assets.
- **Banks.** The definition is broad and can include commercial banks, savings banks, postal banks, credit unions, mortgage associations, and building societies when available.

- **Narrow money.** Narrow money is typically M0 or M1, depending on country and historical availability.
- **Broad money.** Broad money is typically M2 or M3. This is the main monetary aggregate used in comparisons with credit.
- **Real credit growth.** The crisis-prediction analysis frequently uses changes in real bank loans, written as $\Delta \log(\text{loans}/P)$, where P is the price level.

Measurement issue. Historical definitions of banks, money, and credit are not perfectly comparable across countries or over time. The paper therefore emphasizes within-country time-series variation, country fixed effects, and growth rates rather than simple cross-country level comparisons.

3.3 Macroeconomic and asset-price variables

- **Real output.** Real GDP is used to evaluate crisis costs and macroeconomic dynamics.
- **Real investment.** Real investment captures the response of capital formation after crises.
- **Prices.** CPI inflation and the price level are used to study deflationary versus inflationary crisis responses.
- **Stock prices.** Long-run stock market indices are used to test whether asset-price booms improve crisis prediction beyond credit growth.
- **Financial development.** The paper uses loans relative to GDP as a measure of the size of the financial sector in some crisis-prediction specifications.

3.4 Financial-crisis dates

The paper uses an annual coding of systemic banking crises based on historical sources and cross-checks with country experts.

- **Definition.** A crisis is an episode in which the banking sector experiences bank runs, sharp default increases, large capital losses, public intervention, bankruptcy, or forced merger of financial institutions.
- **Sources.** The crisis chronology draws on Bordo et al. and Reinhart and Rogoff, with additional checks against other crisis databases and country-specialist feedback.
- **Event count.** The paper identifies 79 major banking crises in the 14-country sample.
- **Purpose.** These crisis dates are used both for event-study windows and for the crisis-prediction regressions.

Interpretation. Crisis dating is necessarily historical and partly judgment-based. The paper's contribution is not just to use existing dates, but to harmonize them into a consistent long-run panel.

4 Methodology and empirical specifications

4.1 Long-run trends and year effects

To summarize the long-run evolution of money and credit, the paper often estimates fixed-effect regressions of the form

$$x_{it} = a_i + b_t + e_{it}, \quad (1)$$

where x_{it} is a financial aggregate or ratio for country i in year t , a_i is a country fixed effect, and b_t is a year effect. The plotted series are the estimated year effects.

- The country effects remove stable level differences across countries.
- The year effects show the common global component of the money-credit relationship.
- This approach is used for ratios such as broad money to GDP, loans to GDP, assets to GDP, loans to broad money, and assets to broad money.

Interpretation. These figures are not causal regressions. They are disciplined historical summaries that reveal a major structural shift: credit and bank assets grow much faster than money after World War II.

4.2 Event analysis after financial crises

The event analysis compares the behavior of financial and real variables around banking crises.

- The authors align countries around crisis year 0.
- They report average dynamics from normal years through crisis years 0–5.
- They compare pre-World War II crises with post-World War II crises.
- They also calculate cumulative log-level deviations over years 0–5 relative to noncrisis trend.

The main variables in the event analysis are broad money, narrow money, bank loans, bank assets, real GDP, real investment, and the price level.

Interpretation. The event analysis asks what happens after crises. It is descriptive but economically revealing. Prewar crises featured money contractions and deflation. Postwar crises were met with stronger monetary responses, so money did not collapse, but credit and real activity still suffered.

4.3 Financial-crisis prediction model

The central early-warning specification predicts a crisis indicator using lagged credit growth. A simplified logit specification is

$$\Pr(\text{Crisis}_{it} = 1) = \Lambda \left(\alpha_i + \sum_{k=1}^5 \beta_k \Delta \log(\text{Loans}/P)_{i,t-k} \right), \quad (2)$$

where $\Lambda(\cdot)$ is the logistic function and α_i is a country fixed effect.

- The paper also reports linear probability and OLS versions of the same idea.
- The benchmark logit with country fixed effects is the preferred specification.
- The five lag coefficients jointly test whether credit growth over the previous five years predicts crises.
- The sum of the lag coefficients captures the cumulative effect of a credit boom.

Interpretation. The core empirical question is not whether credit contracts during a crisis. It is whether rapid credit growth *before* a crisis helps predict the crisis.

4.4 Money versus credit

The paper compares credit growth with alternative monetary and credit measures.

- Replace real loans with real broad money.
- Replace real loans with real narrow money.
- Use real loans normalized by broad money.
- Use nominal loans instead of real loans.
- Estimate prewar and postwar samples separately.

Interpretation. If a pure money view were sufficient, broad or narrow money growth should predict crises about as well as credit growth. The paper finds the opposite: credit is the stronger and more robust predictor.

4.5 Robustness and additional predictors

The paper adds several controls and alternative predictors.

- Lagged real GDP growth, inflation, current account, interest rates, and investment are added to the baseline credit model.
- Stock price growth is added to test whether equity booms contain independent warning information.
- Loans/GDP is added to test whether larger financial systems are more crisis-prone.
- Interaction terms test whether credit booms are more dangerous when combined with asset-price booms or a larger financial sector.

Interpretation. Credit growth remains important across these checks. Asset prices and financial-sector size matter in some specifications, but they do not replace the predictive content of credit booms.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies robust historical patterns: credit booms tend to precede financial crises in advanced economies. This is stronger than a simple statement that credit falls during crises. The predictive regressions use lagged variables, so the timing runs from credit growth to later crisis realization.

Important limitation. The regressions are not a structural causal model proving that credit growth mechanically causes crises. Credit growth can proxy for leverage, optimism, risk-taking, asset-price dynamics, regulatory failure, or a combination of these forces.

5.2 Why the credit variable is informative

Real credit growth is informative because it captures balance-sheet expansion. A banking system can expand lending rapidly even if broad money does not rise one-for-one, especially when banks fund loans with nonmonetary liabilities.

- Before World War II, bank lending and broad money moved together more closely.
- After World War II, bank loans and assets increasingly decoupled from broad money.
- Therefore, money growth becomes a less complete measure of banking-sector expansion in the modern era.

5.3 Why country fixed effects matter

Countries differ in banking systems, financial development, crisis history, institutions, and data definitions. Country fixed effects remove stable differences across countries and focus the prediction exercise on within-country changes in credit growth.

Interpretation. The model asks whether a given country is more likely to have a crisis after its own credit growth has been unusually strong, not simply whether high-credit countries have more crises.

5.4 Forecasting versus explanation

The ROC and AUROC statistics evaluate classification performance. The paper does not choose a single crisis-warning threshold, because a policymaker's optimal cutoff depends on the relative cost of false alarms and missed crises.

- A perfect classifier has AUROC equal to 1.
- A coin toss has AUROC equal to 0.5.
- The baseline model has AUROC around 0.717, so it has predictive power but is far from perfect.

5.5 Postwar policy interpretation

The postwar regime changed crisis outcomes. Central banks expanded base money more aggressively, deposit insurance and bank supervision reduced classic bank-run dynamics, and deflation was largely avoided.

But real costs remained. Postwar crises still involved significant declines in output and investment, and bank assets sometimes contracted sharply. This suggests that preventing monetary collapse is not the same as preventing balance-sheet deleveraging.

5.6 Main interpretation of the paper

The paper's strongest interpretation is that modern macro-financial risk is tied to credit and leverage cycles. The scale and funding structure of bank balance sheets matter independently of narrow and broad money aggregates.

6 Tables and figures

6.1 Table 1 and Figure 1: Summary statistics and the rise of financial aggregates

Read Table 1:

- Table 1 separates pre-World War II and post-World War II summary statistics.
- Postwar broad money, loans, and assets all grow faster on average than in the prewar period.
- The most important contrast is that loan growth and asset growth rise more than broad money growth after World War II.
- Loan-to-money and asset-to-money growth rates are much higher postwar, indicating a structural rise in leverage and nonmonetary bank funding.

Read Figure 1:

- Figure 1 plots bank loans/GDP, bank assets/GDP, and broad money/GDP using year effects.
- Before World War II, these ratios are comparatively stable, apart from the Great Depression.
- After World War II, bank loans and bank assets rise strongly relative to GDP.

Economic message. The table and figure establish the paper's first stylized fact: advanced economies entered a new postwar era in which credit and bank assets grew much faster than the real economy and faster than broad money.

	Pre-World War II			Post-World War II		
	N	Mean	SD	N	Mean	SD
Loans/money	665	0.4217	0.3582	831	0.5470	0.4239
Assets/money	617	0.7132	0.4453	828	1.0135	0.6688
Broad money/GDP	742	0.5343	0.2070	834	0.6458	0.2405
Loans/money	642	0.7581	0.4382	833	0.8380	0.4942
Assets/money	586	1.2790	0.5642	831	1.5758	0.7525
$\Delta \log$ Real GDP	868	0.0148	0.0448	854	0.0270	0.0253
$\Delta \log$ CPI	826	-0.0002	0.0568	852	0.0452	0.0396
$\Delta \log$ Narrow money	787	0.0278	0.0789	825	0.0780	0.0717
$\Delta \log$ Money	741	0.0365	0.0569	833	0.0857	0.0552
$\Delta \log$ Loans	652	0.0416	0.0898	833	0.1094	0.0749
$\Delta \log$ Assets	607	0.0433	0.0691	825	0.1048	0.0678
$\Delta \log$ Loans/money	626	0.0017	0.0729	825	0.0222	0.0643
$\Delta \log$ Assets/money	573	0.0043	0.0452	820	0.0182	0.0595

Notes: Money denotes broad money. Loans denote total bank loans. Assets denote total bank assets. The sample runs from 1870 to 2008. War and aftermath periods are excluded (1914–1919 and 1939–1947), as is the post-WWI German crisis (1920–1925). The 14 countries in the sample are the United States, Canada, Australia, Denmark, France, Germany, Italy, Japan, the Netherlands, Norway, Spain, Sweden, Switzerland, and the United Kingdom.

Table 1: Annual summary statistics by period.

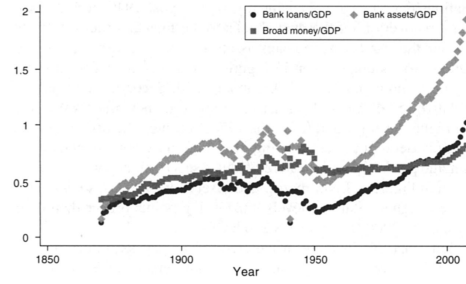


FIGURE 1. AGGREGATES RELATIVE TO GDP (Year Effects)

Figure 1: Aggregates relative to GDP.

6.2 Figures 2 and 3: Credit decouples from broad money

Read Figure 2:

- Figure 2 plots the loan-to-broad-money and asset-to-broad-money ratios.
- Before World War II, the ratios are relatively stable except during the global credit crunch of the 1930s.
- After World War II, loans and assets trend strongly upward relative to broad money.

Read Figure 3:

- Figure 3 shows that the postwar increase in credit relative to money is not confined to one country.
- Many countries display the same upward movement in loans/broad money and assets/broad money.
- This supports the view that the shift is a broad feature of modern finance capitalism rather than a single-country anomaly.

Economic message. Modern banks finance credit creation with more than monetary liabilities. This is why a pure money aggregate becomes an incomplete indicator of financial-system risk.

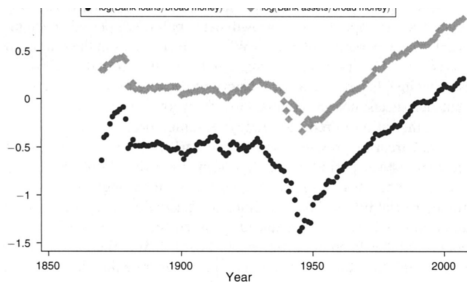


FIGURE 2. AGGREGATES RELATIVE TO BROAD MONEY (Year Effects)

Figure 2: Aggregates relative to broad money.

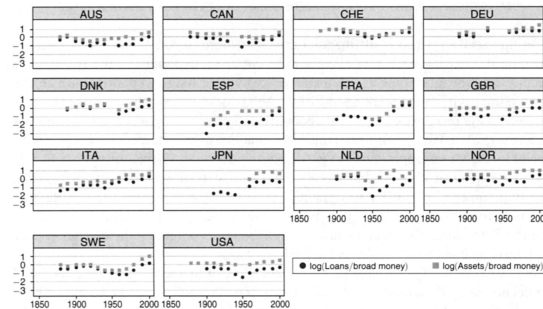


FIGURE 3. AGGREGATES RELATIVE TO BROAD MONEY (By Country)

Figure 3: Aggregates relative to broad money by country.

6.3 Figures 4 and 5: Financial and real variables after crises

Read Figure 4:

- Figure 4 shows the behavior of broad money, narrow money, bank loans, and bank assets around crises.
- Prewar crises are followed by visible weakness in money and credit growth.
- Postwar crises show less collapse in money aggregates, consistent with stronger monetary-policy responses.
- Bank credit and bank assets still weaken after crises, even when money is supported.

Read Figure 5:

- Figure 5 shows real GDP and real investment around crises.
- Crises have large real costs in both eras.
- The postwar policy regime does not eliminate output and investment losses.

Economic message. Avoiding a monetary collapse helps prevent debt deflation, but it does not fully offset the real effects of banking-sector deleveraging.

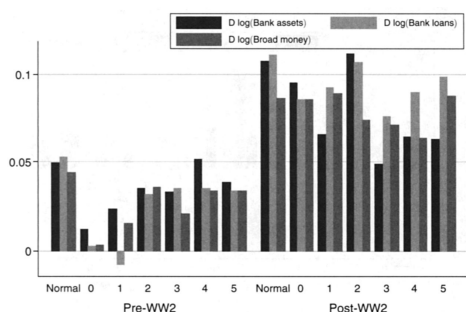


FIGURE 4. AGGREGATES (Postcrisis Periods Relative to Normal)

Figure 4: Money, credit, and bank balance sheets after crises.

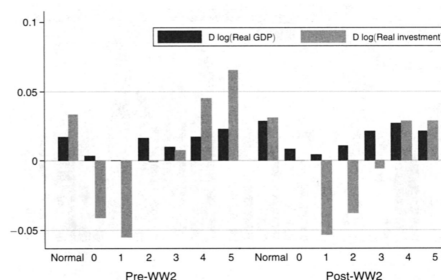


FIGURE 5. REAL VARIABLES (Postcrisis Periods Relative to Normal)

Figure 5: Real GDP and real investment after crises.

6.4 Figure 6 and Table 2: Inflation and cumulative crisis effects

Read Figure 6:

- Figure 6 shows broad money, narrow money, and CPI inflation around crises.
- Prewar crises often involve deflationary pressure.
- Postwar crises tend to avoid sustained deflation, consistent with more aggressive monetary-policy responses and different monetary institutions.

Read Table 2:

- Table 2 reports cumulative log-level effects over crisis years 0–5 relative to noncrisis trend.

- Before World War II, broad money, narrow money, bank loans, bank assets, and the price level fall below trend after crises.
- After World War II, the price-level effect is not negative, but bank loans, bank assets, real GDP, and real investment remain significantly below trend.
- The postwar regime changes the nominal response more clearly than the real cost of crises.

Economic message. The postwar policy regime learned one major lesson from the Great Depression: prevent monetary collapse and deflation. But credit contraction and real recession remain central features of financial crises.

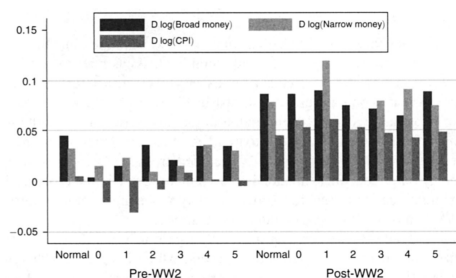


FIGURE 6. MONEY AND INFLATION (Postcrisis Periods Relative to Normal)

Figure 6: Money and inflation after crises.

Cumulative log level effect, after years 0–5 of crisis, versus noncrisis trend, for:	Pre-World War II	Pre-World War II, excluding 1930s	Post-World War II
Log broad money	–0.139*** (0.027)	–0.103*** (0.029)	–0.077** (0.040)
Log narrow money	–0.083** (0.037)	–0.098*** (0.036)	0.009 (0.053)
Log bank loans	–0.248*** (0.044)	–0.220*** (0.047)	–0.144*** (0.055)
Log bank assets	–0.156*** (0.035)	–0.144*** (0.038)	–0.258*** (0.050)
Log real GDP	–0.041** (0.020)	–0.018 (0.020)	–0.079*** (0.018)
Log real investment	–0.190** (0.091)	–0.115 (0.089)	–0.257*** (0.049)
Log price level	–0.089*** (0.025)	–0.055*** (0.026)	0.007 (0.029)

Note: Standard errors in parentheses.
 ***Significant at the 1 percent level.
 **Significant at the 5 percent level.
 *Significant at the 10 percent level.

Table 2: Cumulative effects after financial crises.

6.5 Table 3 and Figure 7: Baseline crisis prediction

Read Table 3:

- Table 3 estimates crisis-prediction models using five lags of real loan growth.
- The second lag of real loan growth is large and statistically significant across several specifications.
- The sum of lag coefficients is positive and significant, meaning that a credit boom over the previous five years predicts a higher crisis probability.
- The preferred logit model with country fixed effects has meaningful predictive ability.

Read Figure 7:

- Figure 7 plots the ROC curve for the baseline model.
- The curve lies above the 45-degree line, meaning the model classifies crises better than a coin toss.
- The reported AUROC is about 0.717, indicating useful but imperfect early-warning power.

Economic message. Credit booms contain early-warning information. The model is not a mechanical crisis alarm, but it shows that ignoring credit growth means discarding a statistically meaningful signal.

Estimation method	OLS	OLS	OLS	Logit	Logit
Fixed effects	None	Country	Country + year	None	Country
Specification	(1)	(2)	(3)	(4)	Baseline
L. Δ log (loans/P)	-0.0281 (0.0812)	-0.0273 (0.0815)	-0.0489 (0.0801)	-0.257 (2.077)	-0.398 (2.110)
L2. Δ log (loans/P)	0.301*** (0.0869)	0.302*** (0.0872)	0.320*** (0.0833)	6.956*** (2.308)	7.138*** (2.631)
L3. Δ log (loans/P)	0.0486 (0.0850)	0.0478 (0.0853)	0.00134 (0.0819)	1.079 (2.826)	0.888 (2.948)
L4. Δ log (loans/P)	0.00494 (0.0811)	0.00213 (0.0814)	0.0346 (0.0782)	0.290 (1.282)	0.203 (1.378)
L5. Δ log (loans/P)	0.0979 (0.0746)	0.0928 (0.0752)	0.136 (0.0729)	2.035 (1.607)	1.867 (1.640)
Observations	1,272	1,272	1,272	1,272	1,272
Groups	14	14	14	14	14
Sum of lag coefficients	0.425***	0.417***	0.443***	10.10***	9.697***
Standard error	0.123	0.126	0.136	2.590	2.920
Test for all lags = 0 ⁱ	4.061***	3.871***	4.328***	24.95***	17.23***
p-value	0.00116	0.00174	0.000661	0.000143	0.00408
Test for country effects = 0 ^j	—	0.71	0.84	—	7.67
p-value	—	0.754	0.617	—	0.864
Test for year effects = 0 ^j	—	—	4.15***	—	—
p-value	—	—	0.0001	—	—
R ² ⁱⁱ	0.016	0.023	0.290	0.0434	0.0659
Pseudolikelihood	—	—	—	-210.8	-205.8
Overall test statistics ⁱ	4.061***	1.638*	4.184***	24.95***	36.21***
p-value	0.0012	0.0445	0.00001	0.000143	0.00663
AUROC	0.673***	0.720***	0.952***	0.673***	0.717***
Standard error	0.0357	0.0341	0.00865	0.0360	0.0349

Notes: ⁱReported statistic is F for OLS, χ^2 for logit. ⁱⁱReported statistic is Pseudo R^2 for logit. Standard errors in parentheses. Logit standard errors are robust.
***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

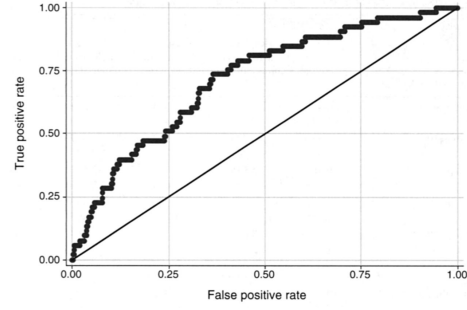


FIGURE 7. RECEIVER OPERATING CHARACTERISTIC CURVE (Baseline Model)

Note: Area under ROC curve = 0.717.

Figure 7: ROC curve for the baseline model.

Table 3: Financial crisis prediction using lagged real loan growth.

6.6 Tables 4 and 5: Credit versus money, and prewar versus postwar samples

Read Table 4:

- Table 4 replaces the baseline real-loan variable with alternative money and credit measures.
- Broad money and narrow money perform worse than real bank loans.
- Real loans normalized by broad money still retain predictive content, reinforcing the idea that credit contains information beyond money.
- The AUROC comparisons generally favor credit over money.

Read Table 5:

- Table 5 estimates the baseline model separately before and after World War II.
- Credit growth predicts crises in both periods.
- Money growth performs much less well in the postwar period.
- This pattern fits the historical evidence that credit and money decouple after World War II.

Economic message. The main early-warning result is not a disguised money-growth result. The signal is specifically tied to bank credit and balance-sheet expansion.

TABLE 4—BASELINE MODEL AND ALTERNATIVE MEASURES OF MONEY AND CREDIT

Specification (Logit country effects)	Baseline (6)	Replace loans with broad money (7)	Replace loans with narrow money (8)	Replace real loans with loans/GDP (9)	Replace real loans with loans/broad money (10)
L. Δ log (loans/P)	-0.398	-1.051	-2.504	2.091	0.601
	-2.11	-2.771	-1.806	-2.235	-2.383
L2. Δ log (loans/P)	7.138***	5.773***	2.303	7.627***	5.842**
	-2.631	-2.181	-1.781	-2.135	-2.327
L3. Δ log (loans/P)	0.888	3.515	1.768	3.569	2.092
	-2.948	-2.329	-1.664	-2.386	-2.048
L4. Δ log (loans/P)	0.203	-1.535	-2.880*	2.333*	1.613
	-1.378	-2.287	-1.51	-1.405	-1.766
L5. Δ log (loans/P)	1.867	3.077	1.373	3.164**	0.497
	-1.64	-2.256	-1.63	-1.583	-2.37
Marginal effects at each lag	-0.0124	-0.0350	-0.0888	0.0598	0.0196
evaluated at the means	0.222	0.192	0.0817	0.218	0.190
	0.0276	0.117	0.0627	0.102	0.0681
	0.00629	-0.0511	-0.102	0.0668	0.0525
	0.0580	0.102	0.0487	0.0905	0.0162
Sum	0.301	0.326	0.00211	0.538	0.346
Observations	1,272	1,348	1,381	1,245	1,224
Groups	14	14	14	14	14
Sum of lag coefficients	9.697***	9.779***	0.0596	18.78***	10.65***
Standard error	2.920	3.400	3.240	3.651	4.053
Test for all lags = 0, χ^2	17.23***	17.77***	6.557	29.85***	10.62*
p-value	0.00408	0.00324	0.256	0.000016	0.0594
Test for country effects = 0, χ^2	7.674	8.755	8.834	8.012	9.140
p-value	0.864	0.791	0.785	0.843	0.762
Pseudo R^2	0.0659	0.0487	0.0381	0.0923	0.0497
Pseudolikelihood	-205.8	-224.6	-237.4	-198.9	-201.5
Overall test statistic, χ^2	36.21***	36.81***	17.37	47.77***	19.82
p-value	0.00663	0.00555	0.498	0.000163	0.343
Predictive ability, AUROC	0.717***	0.681***	0.631***	0.743***	0.680***
Standard error	0.0349	0.0294	0.0339	0.0337	0.0378

Note: Robust standard errors in parentheses.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

TABLE 5—BASELINE MODEL WITH PRE-WW2 AND POST-WW2 SAMPLES

Specification (Logit country effects)	Baseline pre-WW2 sample using loans (11)	Baseline post-WW2 sample using loans (12)	Pre-WW2 sample replace loans with broad money (13)	Post-WW2 sample replace loans with broad money (14)
L. Δ log (loans/P)	2.249	-0.316	-0.227	2.705
	(2.362)	(3.005)	(3.014)	(4.438)
L2. Δ log (loans/P)	7.697**	8.307***	7.293**	4.719**
	(3.221)	(2.497)	(3.004)	(2.375)
L3. Δ log (loans/P)	2.890	2.946	4.077	4.060*
	(3.056)	(2.687)	(2.915)	(2.170)
L4. Δ log (loans/P)	2.486	0.755	-0.249	-0.838
	(1.587)	(2.623)	(1.982)	(5.359)
L5. Δ log (loans/P)	4.260**	-1.749	4.844*	0.808
	(1.735)	(3.204)	(2.647)	(4.016)
Observations	510	706	585	708
Groups	13	14	13	14
Marginal effects at each lag	0.0873	-0.00642	-0.0102	0.0617
evaluated at the means	0.299	0.169	0.332	0.108
	0.112	0.0598	0.183	0.0926
	0.0965	0.0153	-0.0112	-0.0191
	0.165	-0.0355	0.218	0.0184
Sum	0.760	0.202	0.711	0.261
Sum of lag coefficients	19.58***	9.943	15.84***	11.45*
Standard error	4.921	6.056	5.119	6.022
Test for all lags = 0, χ^2	19.20***	12.44**	13.53**	12.13**
p-value	0.00176	0.0292	0.0189	0.0330
Test for country effects = 0, χ^2	6.369	5.348	11.74	5.917
p-value	0.932	0.945	0.549	0.920
Pseudo R^2	0.130	0.0771	0.0855	0.0476
Pseudolikelihood	-106.4	-83.97	-126.2	-86.71
Overall test statistic, χ^2	40.21***	36.44***	35.95***	19.89
p-value	0.00195	0.00401	0.00716	0.280
AUROC	0.763***	0.718***	0.728***	0.659***
Standard error	0.0391	0.0691	0.0361	0.0600

Notes: Robust standard errors in parentheses. In the prewar sample NLD is dropped from the logit regression because there are no crises in the sample (with five lags of credit or money in nonwar years), so $N = 13$ for these cases.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

Table 4: Baseline model and alternative measures of money and credit.

Table 5: Prewar and postwar crisis-prediction samples.

6.7 Figure 8 and Table 6: Predictive comparisons and robustness checks

Read Figure 8:

- Figure 8 compares ROC curves for money and credit predictors in prewar and postwar subsamples.
- In the prewar sample, credit and money look more similar.
- In the postwar sample, credit outperforms money more clearly.
- The figure visually reinforces the paper's two-era argument.

Read Table 6:

- Table 6 adds lags of real GDP growth, inflation, current account, interest rates, and investment to the baseline logit model.
- The lagged real-loan coefficients remain economically important across these robustness checks.
- Added macro variables do not eliminate the predictive power of credit growth.

Economic message. Credit growth is not merely proxying for normal business-cycle conditions or standard macro controls. It remains a distinct crisis-warning variable.

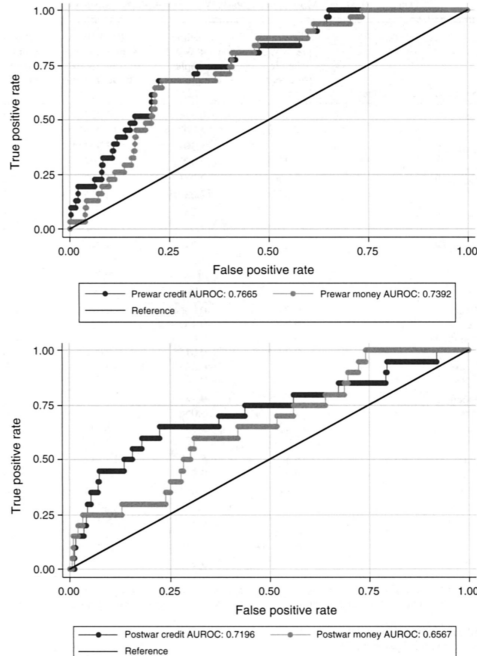


Figure 8: ROC comparisons of money and credit predictors.

Specification (Logit country effects)	Baseline plus 5 lags of real GDP growth (15)	Baseline plus 5 lags of inflation (16)	Baseline plus 5 lags of nominal short- term int. rate (17)	Baseline plus 5 lags of real short- term int. rate (18)	Baseline plus 5 lags of change in I/Y (19)
L. $\Delta \log$ (loans/P)	1.544 (2.081)	-0.771 (2.362)	0.113 (2.072)	-1.458 (2.442)	-0.152 (2.250)
L2. $\Delta \log$ (loans/P)	8.571*** (2.403)	10.22*** (2.690)	8.491*** (2.982)	10.99*** (2.689)	7.131*** (2.845)
L3. $\Delta \log$ (loans/P)	3.114 (2.347)	-1.590 (2.856)	1.054 (3.443)	-1.258 (3.239)	0.546 (3.213)
L4. $\Delta \log$ (loans/P)	2.555* (1.441)	1.503 (1.461)	0.241 (1.697)	2.686 (1.673)	1.124 (1.366)
L5. $\Delta \log$ (loans/P)	3.539*** (1.555)	1.128 (1.842)	1.996 (2.058)	0.821 (1.820)	3.004 (1.943)
Observations	1,272	1,272	969	961	1,218
Groups	14	14	14	14	14
Sum of lag coefficients	19.32***	10.49***	11.89***	11.78***	11.65***
Standard error	4.329	3.121	3.275	3.385	3.404
Test for all lags = 0, χ^2	28.99***	24.45***	19.45***	26.54	16.59
p-value	0.000002	0.000178	0.00159	0.000007	0.00536
Test lags of added vbl. = 0, χ^2	34.43	19.55	3.888	14.98	4.895
p-value	0.000002	0.00152	0.566	0.0105	0.429
Test for country effects = 0, χ^2	10.41	8.031	7.727	5.888	8.538
p-value	0.660	0.842	0.806	0.922	0.807
Pseudo R^2	0.123	0.104	0.0765	0.110	0.0870
Pseudolikelihood	-193.3	-197.4	-181.8	-174.9	-196.2
Overall test statistic, χ^2	61.47***	52.23***	36.40**	47.46***	52.38***
p-value	0.00023	0.000466	0.0275	0.00128	0.000445
AUROC	0.751***	0.770***	0.725***	0.750***	0.737***
Standard error	0.0452	0.0422	0.0448	0.0454	0.0419

Notes: Robust standard errors in parentheses. Marginal effects not reported.
***Coefficient at the 1 percent level

Table 6: Robustness checks with additional macro controls.

6.8 Tables 7 and 8: Asset prices, financial development, and interactions

Read Table 7:

- Table 7 adds stock price growth and financial development measures to the baseline model.
- Credit growth remains significant in specifications with stock-price controls.
- Stock price booms contain some information, but they do not subsume the credit signal.
- Loans/GDP is positive in some specifications, consistent with the idea that larger financial systems can be more fragile.

Read Table 8:

- Table 8 uses five-year moving averages and interaction terms involving credit growth, stock price growth, and financial development.
- The specifications ask whether credit booms are especially dangerous when they occur alongside asset-price booms or in more financialized economies.
- The interaction evidence is less decisive than the baseline credit-growth result.
- The main conclusion remains: accelerations in credit creation are the prime suspect in crisis prediction.

Economic message. Asset prices and financial development can matter, but the cleanest and most robust signal is credit growth. Credit booms should therefore be central to any macro-financial surveillance framework.

TABLE 7—CREDIT, ASSET PRICES, AND FINANCIAL DEVELOPMENT

Specification (Logit country effects)	Baseline plus 5 lags of nominal stock price change (20)	Baseline plus 5 lags of real stock price change (21)	Baseline plus Loans over GDP (22)	Baseline plus 5 lags of real stock prices plus loans/GDP (23)
L. Δ log (loans/P)	-2.491 (2.324)	-2.540 (2.312)	-0.755 (2.293)	-3.392 (2.470)
L2. Δ log (loans/P)	7.316** (2.910)	7.165** (2.915)	7.599*** (2.871)	7.848** (3.215)
L3. Δ log (loans/P)	3.405 (2.899)	3.185 (2.864)	0.720 (3.307)	3.297 (3.171)
L4. Δ log (loans/P)	-1.352 (1.521)	-1.684 (1.539)	0.0933 (1.497)	-1.747 (1.669)
L5. Δ log (loans/P)	1.678 (1.835)	1.771 (1.784)	2.326 (1.784)	2.460 (1.994)
L1. Δ log (stock prices)	-1.046** (0.464)	-0.865** (0.434)	-0.768* (0.455)	-0.758* (0.455)
L2. Δ log (stock prices)	0.535 (0.644)	0.563 (0.673)	0.550 (0.666)	0.550 (0.666)
L3. Δ log (stock prices)	0.272 (0.651)	0.715 (0.692)	0.691 (0.690)	0.691 (0.690)
L4. Δ log (stock prices)	0.954 (0.822)	1.098 (0.811)	1.024 (0.814)	1.024 (0.814)
L5. Δ log (stock prices)	0.0844 (0.631)	0.467 (0.703)	0.438 (0.627)	0.438 (0.627)
Loans/GDP (log)			1.100* (0.624)	1.601** (0.703)
Observations	1,061	1,062	1,271	1,061
Groups	14	14	14	14
Sum of lag coefficients	8.557**	7.898**	9.984***	8.466**
Standard error	3.468	3.443	2.918	3.460
Test for all lags = 0, χ^2	22.04***	20.65***	17.45***	21.19***
p-value	0.000515	0.000944	0.00371	0.000747
Test lags of added vbl. = 0, χ^2	8.664	13.28**		11.89**
p-value	0.123	0.0209		0.0363
Test for country effects = 0, χ^2	5.499	5.433	11.43	10.33
p-value	0.939	0.942	0.575	0.587
Pseudo R^2	0.0882	0.0901	0.0749	0.108
Pseudolikelihood	-169.8	-169.5	-203.8	-166.2
Overall test statistic, χ^2	39.65**	46.84***	41.48***	47.20***
p-value	0.0119	0.00154	0.00208	0.00212
AUROC	0.727***	0.731***	0.731***	0.764***
Standard error	0.0399	0.0383	0.0379	0.0358

Table 7: Credit, asset prices, and financial development.

TABLE 8—CREDIT, ASSET PRICES, AND FINANCIAL DEVELOPMENT—INTERACTIONS

Specification (Logit country effects) 5-year moving average of:	Baseline (24)	Baseline plus (25)	Baseline plus (26)	Baseline plus (27)	Baseline plus (28)	Baseline plus (29)
Δ log (loans/P)	5.340*** (2.069)	5.012** (2.288)	7.526*** (2.464)	6.752*** (2.012)	6.632*** (2.243)	7.370*** (2.368)
Δ log (stocks/P)		0.524 (1.391)	2.704 (2.103)			0.236 (1.464)
Δ log (loans/P) $\times$ Δ log (stocks/P)			-22.77 (14.19)			
Loans/GDP (log)				1.432*** (0.530)	1.515** (0.751)	1.704*** (0.615)
Δ log (loans/P) $\times$ loans/GDP (log)					-1.243 (8.516)	
Δ log (stocks/P) $\times$ loans/GDP (log)						4.661 (3.401)
Observations	1,278	1,278	1,278	1,437	1,437	1,278
Groups	14	14	14	14	14	14
Test for country effects = 0, χ^2	7.447	7.526	7.980	15.58	14.58	16.21
p-value	0.878	0.873	0.845	0.272	0.334	0.238
Pseudo R^2	0.0348	0.0351	0.0407	0.0456	0.0457	0.0652
Pseudolikelihood	-203.7	-203.7	-202.5	-246.8	-246.8	-197.3
Overall test statistic, χ^2	17.58	18.47	22.95	35.78***	35.71***	30.92**
p-value	0.227	0.239	0.115	0.00190	0.00317	0.0205
AUROC	0.663***	0.662***	0.669***	0.689***	0.689***	0.714***
Standard error	0.0392	0.0385	0.0369	0.0368	0.0367	0.0371

Notes: Robust standard errors in parentheses. Marginal effects not reported.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

Table 8: Interactions among credit, asset prices, and financial development.

7 Short summary

Schularick and Taylor ask whether the long-run history of money, credit, and banking crises supports a money view or a credit view of macro-financial instability. Their answer is that the modern era strongly supports the credit view.

Before World War II, money and credit moved closely together. This is why classical monetarist narratives that emphasize money are understandable for that period. After World War II, however, bank loans and bank assets grew rapidly relative to both GDP and broad money. This decoupling means that monetary aggregates no longer fully capture the expansion of bank balance sheets.

The event analysis shows that postwar policy prevented the most destructive nominal features of prewar crises: broad money collapse, narrow money collapse, and deflation. Yet banking crises still produced large real costs through falling credit, bank assets, GDP, and investment.

The crisis-prediction analysis shows that rapid credit growth over the previous five years predicts financial crises. This result is robust to alternative specifications, prewar and postwar subsamples, macro controls, stock price controls, and financial-development controls. Money growth does not perform as well, especially after World War II.

8 Conclusion

8.1 Core conclusion

The central conclusion is that credit booms are historically important warning signs of financial crises. The paper's title is literal: many financial crises are credit booms gone bust.

8.2 Economic intuition

A banking system can grow in a way that looks stable from the perspective of broad money but risky from the perspective of leverage. If banks expand loans by issuing nonmonetary liabilities or increasing balance-sheet leverage, then credit can boom without a proportional rise in money. This creates fragility that pure money aggregates may miss.

8.3 Incremental contribution revisited

The paper contributes by combining three pieces of evidence:

- a long-run dataset documenting the historical divergence between money and credit;
- event-study evidence showing how crisis dynamics changed across monetary regimes; and
- crisis-prediction evidence showing that credit growth is a better early-warning signal than money growth.

8.4 Implications

The results imply that central banks and regulators should monitor credit growth, leverage, and bank balance-sheet composition. Inflation targeting alone may deliver price stability while allowing financial instability to build in the background.

8.5 Limitations

- The data are annual, so the analysis cannot capture high-frequency credit-market stress.
- Historical bank and money definitions vary across countries.
- Crisis dates are carefully constructed but still partly judgment-based.
- Predictive regressions are not structural causal models.
- The sample is advanced economies, so the conclusions may not directly apply to emerging markets without caution.

8.6 Takeaway

The paper's practical takeaway is simple: watch credit. When bank lending expands rapidly for several years, the probability of a banking crisis rises. Money aggregates alone are not enough in a

world of leveraged financial intermediaries and nonmonetary bank funding.